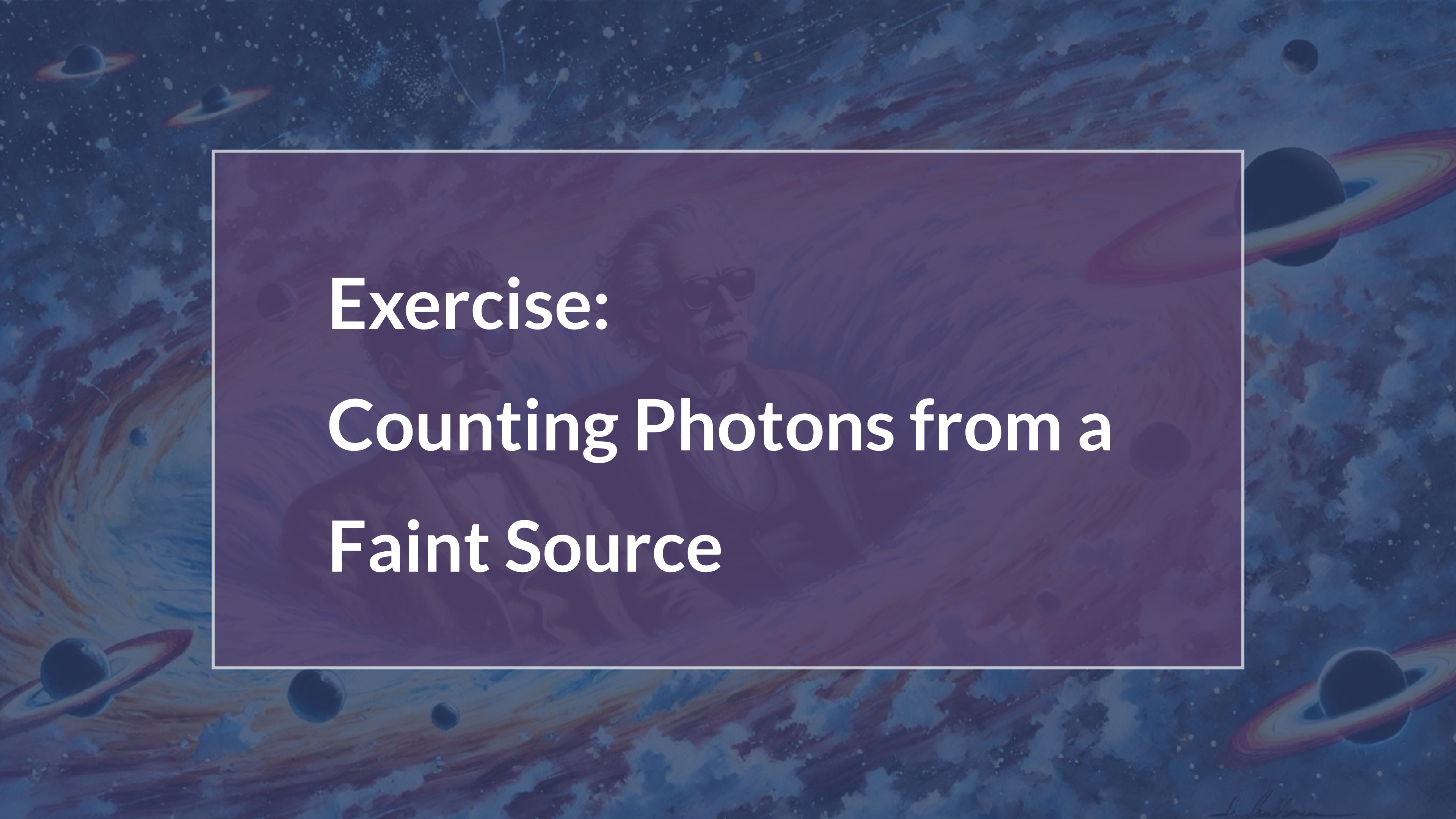


A vibrant, painterly illustration of a cosmic scene. In the center, two men, Albert Einstein and Max Planck, are depicted from the chest up, floating amidst swirling nebulae and galaxies. Einstein, on the right, has his characteristic wild white hair and is wearing dark sunglasses. Planck, on the left, has dark curly hair and a mustache, also wearing sunglasses. They are both dressed in formal attire: Einstein in a dark suit and Planck in a brown jacket over a white shirt and a dark bow tie. The background is a deep space filled with colorful nebulae in shades of blue, purple, and orange, with several ringed planets and smaller celestial bodies scattered throughout. A semi-transparent purple rectangular box is centered over the image, containing the title text.

Distributions

The background of the slide is a vibrant, artistic depiction of outer space. It features swirling galaxies in shades of blue, purple, and orange, interspersed with numerous stars and nebulae. Several planets with prominent ring systems, similar to Saturn, are scattered throughout the scene. In the center, a semi-transparent purple rectangular box contains the text. Within this box, there is a faint, stylized portrait of Albert Einstein, showing him with his characteristic wild hair and mustache, wearing a suit and tie.

Exercise: Counting Photons from a Faint Source

A distribution

... tells you the frequency or relative frequency of each possible value or event (or of some data that was collected)

... can be useful for modelling a population of objects

... is a foundation of statistical reasoning

... can be empirical or analytic

... can be continuous or discrete

... can be univariate or multivariate

Analytic distributions have parameters that define its shape



A probability distribution

... characterises random variables

... tells you how probability is distributed across the possible values of a random variable

... integrates to 1

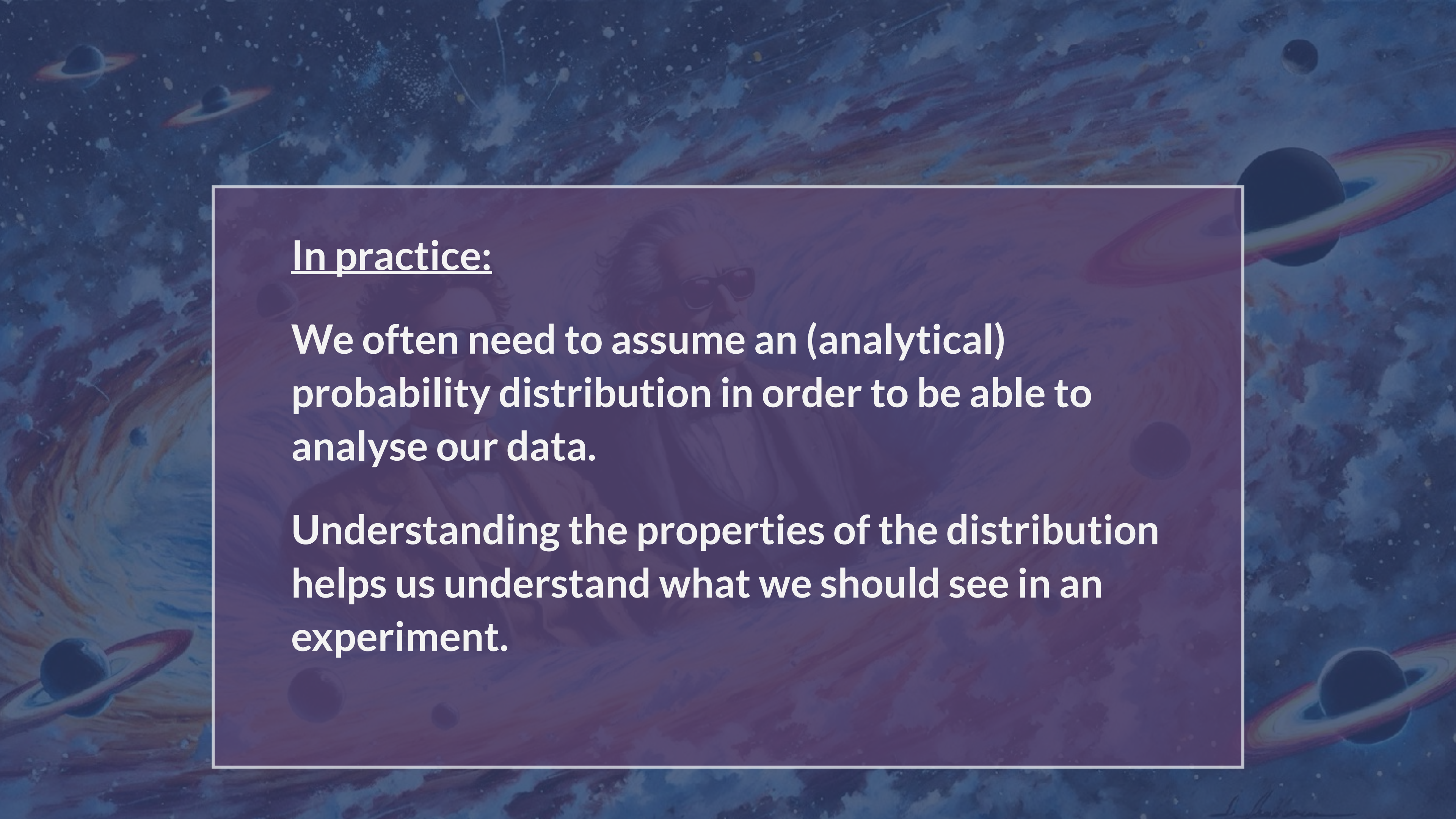
... can be useful for modelling a population, your belief in the possible values of a model parameter, etc.

... can be continuous or discrete

... can be univariate or multivariate

... has parameters that define its shape



The background of the slide is a vibrant, artistic depiction of outer space. It features swirling galaxies in shades of blue, purple, and orange, interspersed with various celestial bodies. Several planets with prominent rings, similar to Saturn, are visible, along with smaller moons and distant stars. The overall aesthetic is that of a classic science fiction or space exploration theme.

In practice:

We often need to assume an (analytical) probability distribution in order to be able to analyse our data.

Understanding the properties of the distribution helps us understand what we should see in an experiment.

Random variables

- Random variables are functions that assign a numerical value to the outcome/event from an experiment
- Notation:
- Map from sample space (Ω) to the real line: $X(\omega)$
- Examples of random variables:



Discrete random variables

Continuous random variables



Discrete random variables

- Example: a coin flip

Continuous random variables



Discrete random variables

- Example: a coin flip
- Sample space:

Continuous random variables



Discrete random variables

- Example: a coin flip
- Sample space:
- Define $X : \Omega \rightarrow \mathbb{R}$ as

Continuous random variables



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- Then the probability for each possible value of X is

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Discrete random variables

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Continuous random variables

- Example: the height of a person

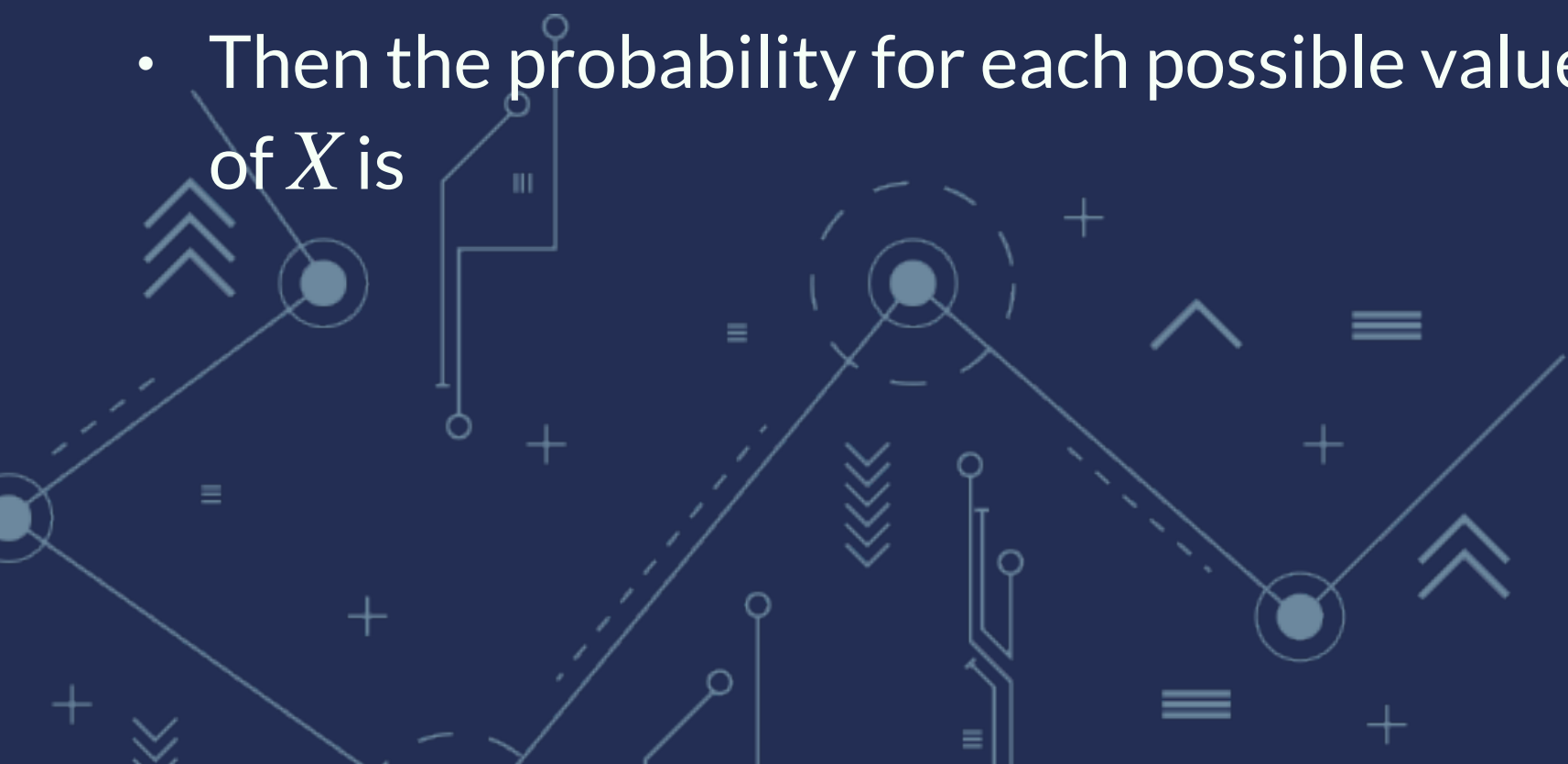


Discrete random variables

- Example: a coin flip
- Sample space:
- Define $X : \Omega \rightarrow \mathbb{R}$ as
- Then the probability for each possible value of X is

Continuous random variables

- Example: the height of a person
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Discrete random variables

- Example: a coin flip
- Sample space:
- Define $X : \Omega \rightarrow \mathbb{R}$ as
- Then the probability for each possible value of X is

Continuous random variables

- Example: the height of a person
- Sample space:
- We can say $X : \Omega \rightarrow \mathbb{R}$ and $X > 0$



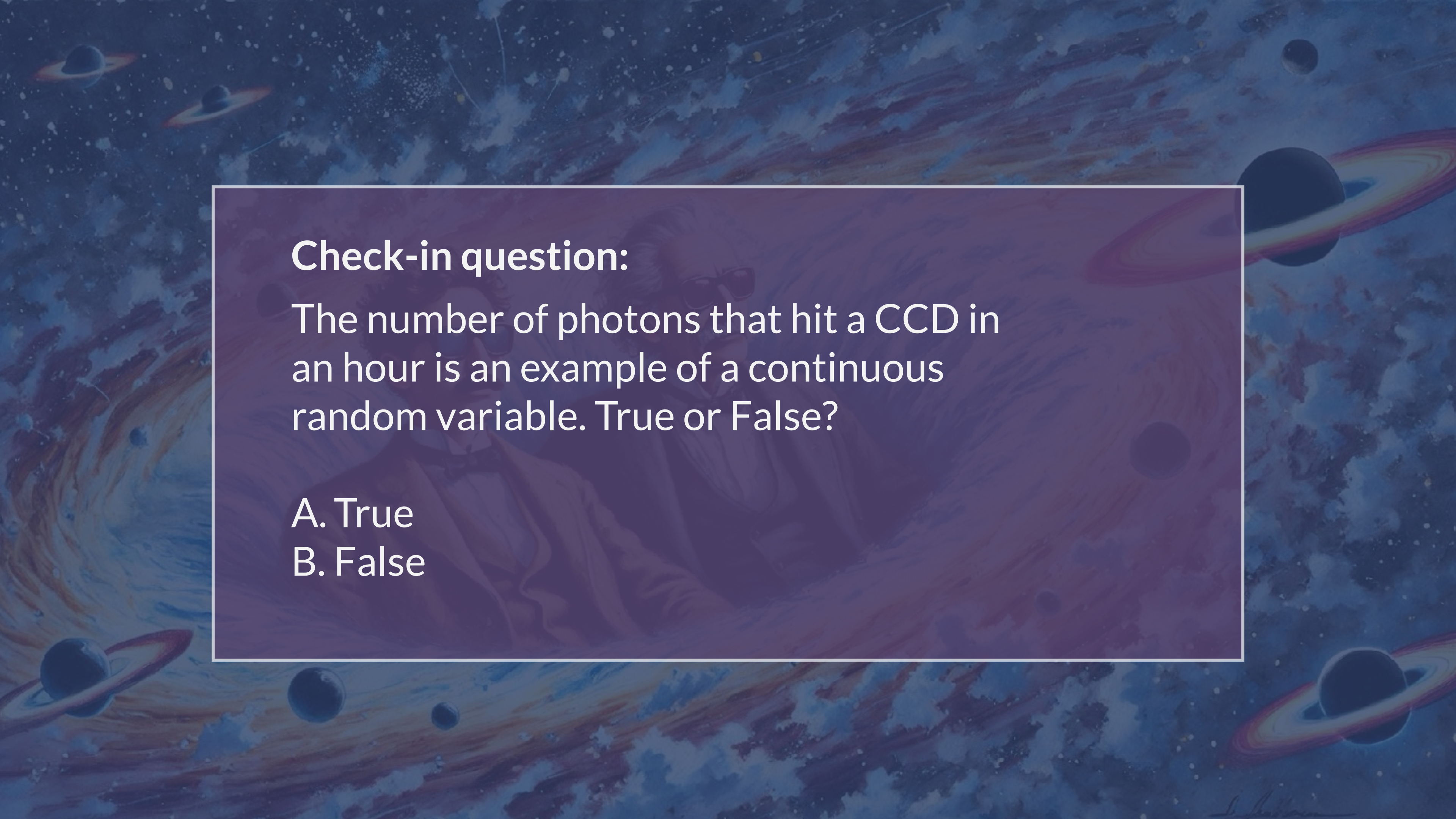
Discrete random variables

- Example: a coin flip
- Sample space:
- Define $X : \Omega \rightarrow \mathbb{R}$ as
- Then the probability for each possible value of X is

Continuous random variables

- Example: the height of a person
- Sample space:
- We can say $X : \Omega \rightarrow \mathbb{R}$ and $X > 0$
- Then the probability for each possible value of X is defined within an interval $P(x \in I)$



The background of the slide is a vibrant, artistic depiction of outer space. It features swirling galaxies in shades of blue, purple, and orange, interspersed with numerous stars and several ringed planets, similar to Saturn, in various orientations. A semi-transparent purple rectangle is centered on the slide, containing the text.

Check-in question:

The number of photons that hit a CCD in an hour is an example of a continuous random variable. True or False?

- A. True
- B. False

Continuous versus discrete probability distributions

Discrete quantities: probability mass function (pmf)

Continuous quantities: probability density function (pdf)



Continuous versus discrete probability distributions

What are some examples of continuous random variables in physics and astronomy?

What are some examples of discrete random variables in astronomy?



Data are
realisations of
random variables

ADD EXAMPLE HERE!



Population

Sample

Statistics can be used to understand the underlying population, when all you have is a sample

Beware: representativeness matters a lot!

Population

Sample

- True, underlying distribution for some quantity

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- e.g. the distribution of movies watched per month across the whole world)

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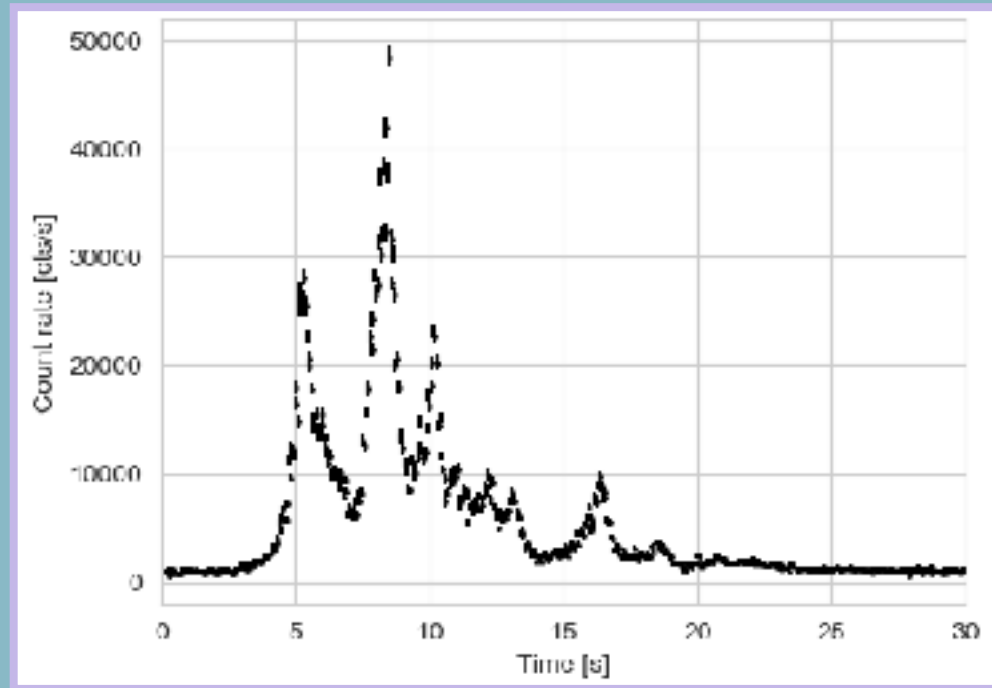
Sample

- A sample drawn from some distribution
- e.g. the Physics and Astronomy MSc students in Amsterdam, in January
- Will not be exactly like the population because of randomness

Statistics can be used to understand the underlying population, when all you have is a sample

Beware: representativeness matters a lot!

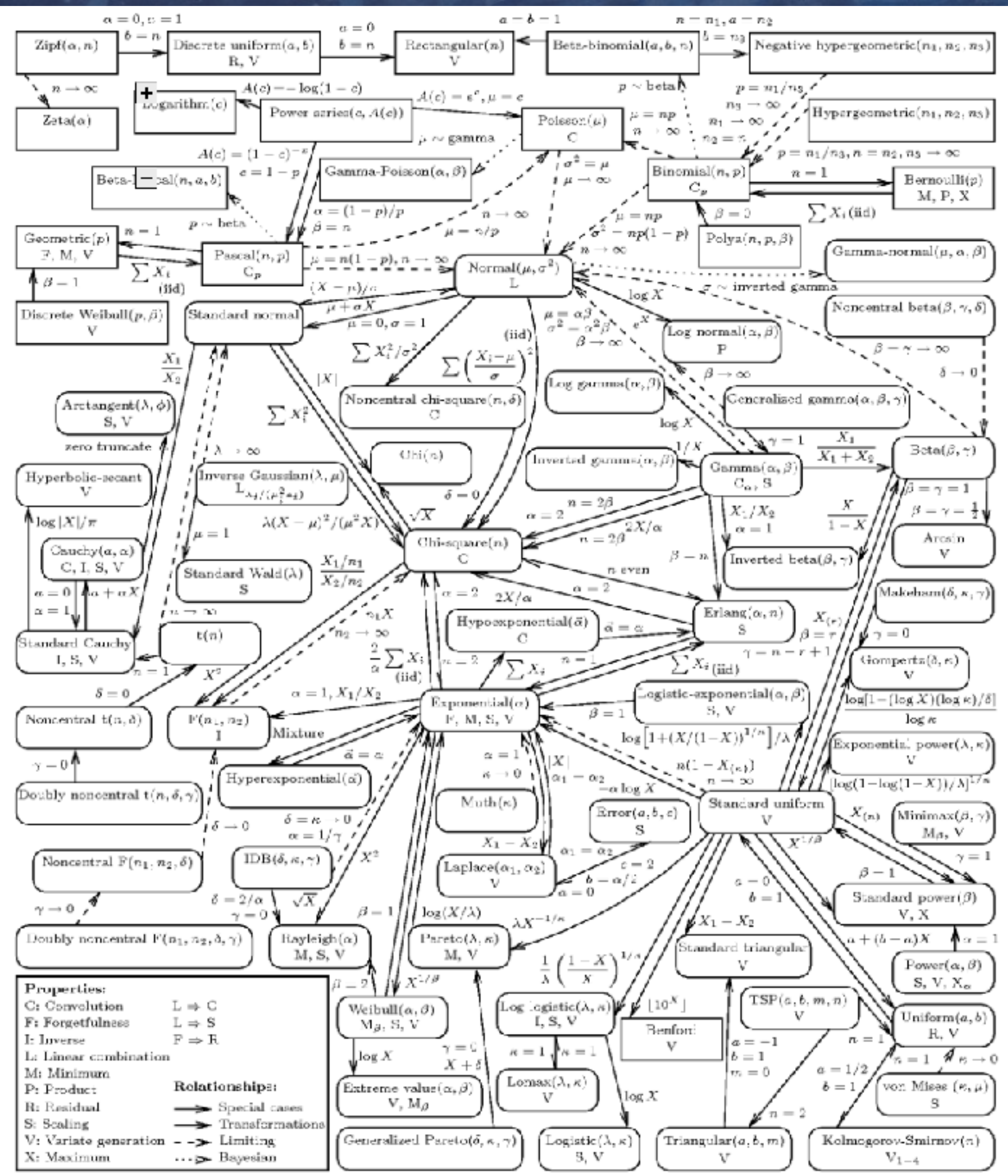
Randomness matters!



- We often don't know how the random variable X is distributed
- Our random data sample x is just that: a sample
- Randomness can trick us! Humans look for patterns in data (pareidolia)
- Things are tricky when data suffer from selection biases, observation bias etc.
- Look at data and summarise it in different ways. Be skeptical!



There are many univariate distributions



What probability
distributions do you
know?



Common Distributions

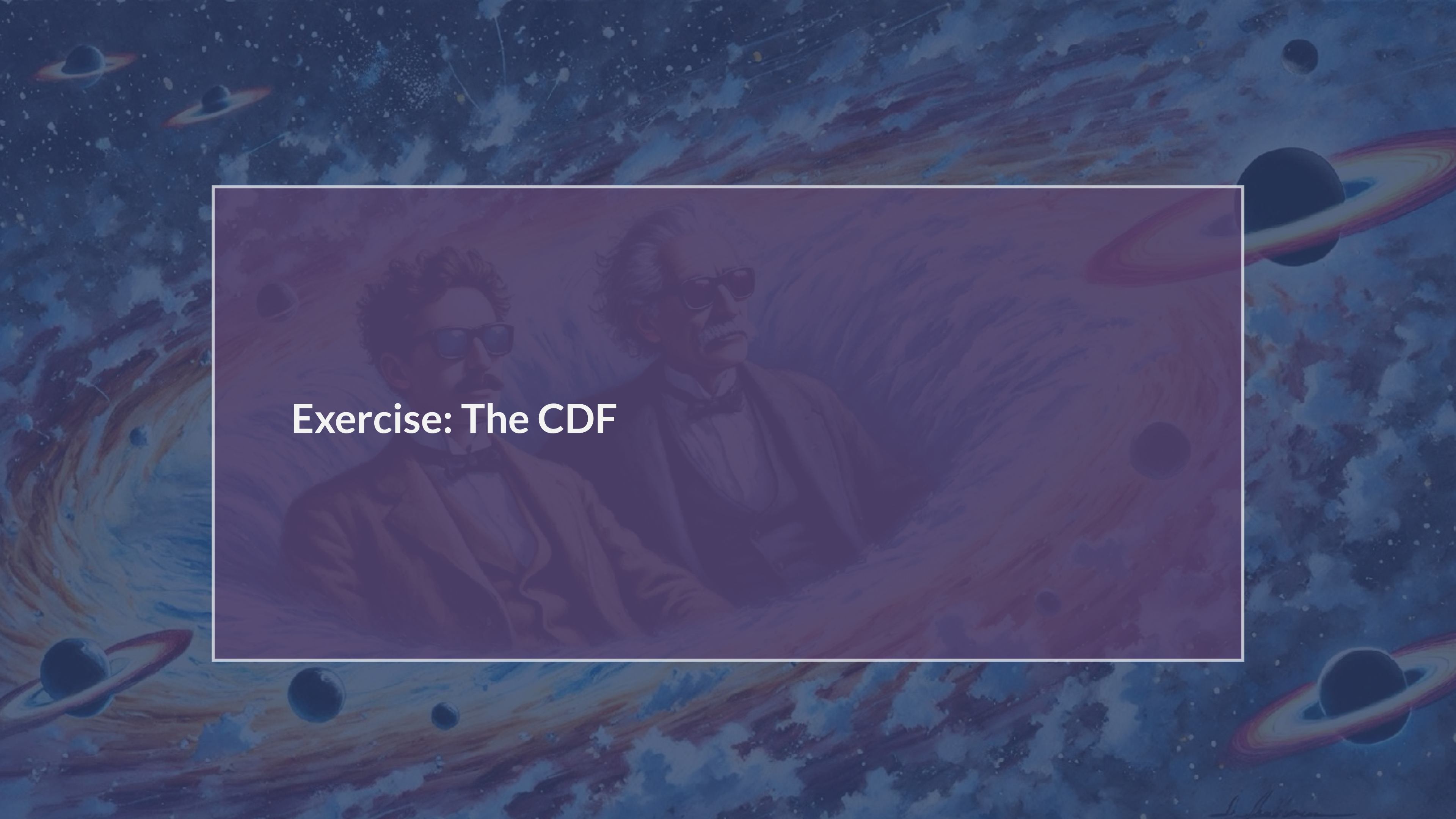
- Bernoulli
 - For binary random variables. Probability of success is p
- Binomial
 - For a sum of Bernoulli n trials, each with probability p
- Poisson
 - For number of events/counts, which happen at some mean rate λ
- Gaussian
 - Naturally arises out of the Central Limit Theorem. Parameters are its mean μ and its variance σ^2
- Student t
 - Used in hypothesis testing for sample means, shape determined by degrees of freedom
- Log-normal
 - When $\log(X)$ is normally distributed, parameters mean μ and its variance σ^2
- Chi-Square
 - When X is the sum of squares of k normally distributed, independent random variables
 - Shape depends on degrees of freedom k

The Normal (Gaussian) Distribution

$$N(\mu, \sigma^2)$$

pdf: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{(x - \mu)^2}{2\sigma^2}\right)$



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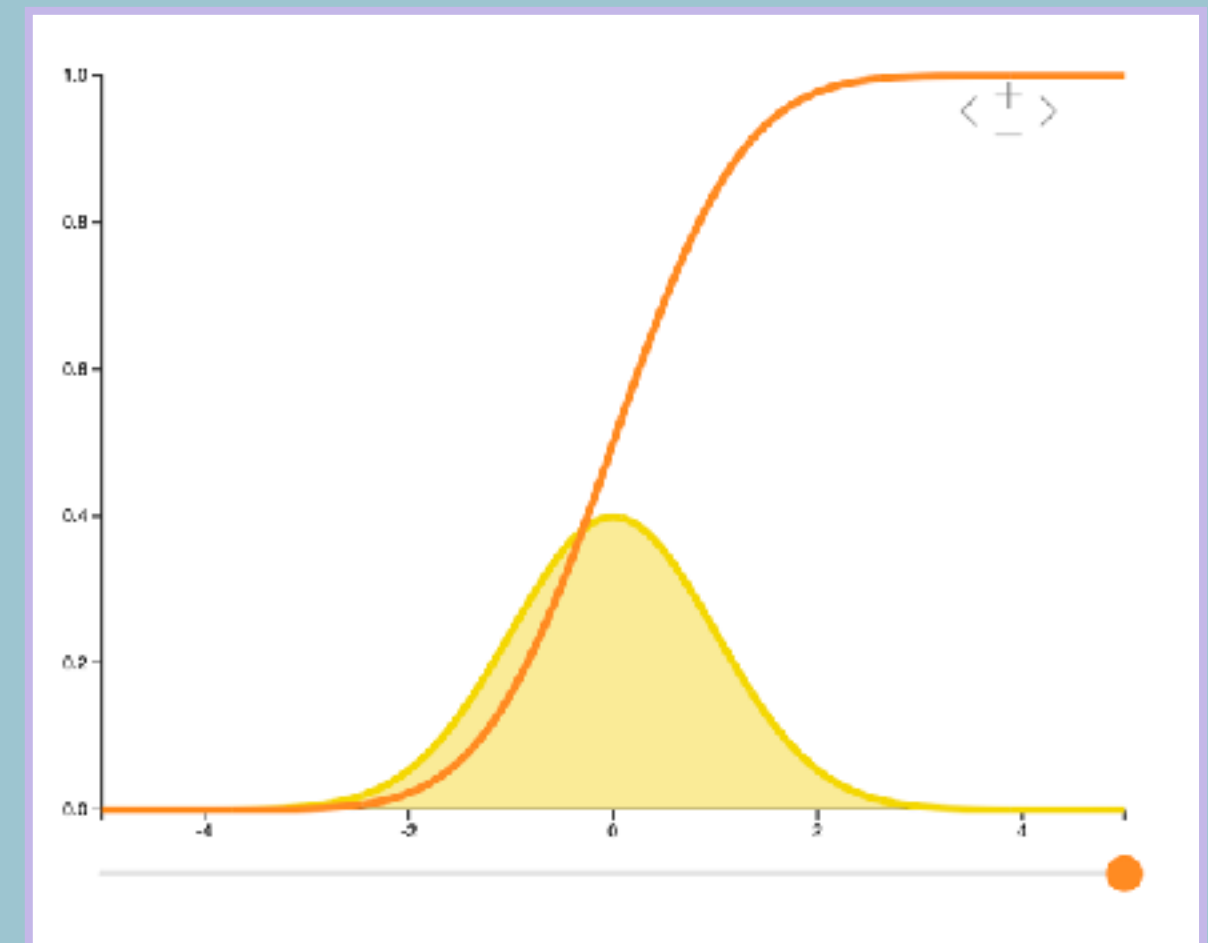
Exercise: The CDF

The Cumulative Distribution Function

$$F_X(x) = P(X \leq x)$$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$



Percentiles

Quantile: split the distribution into segments of equal probability

Quartiles: 25%, 50%, 75% quantiles

Percentiles: split distribution into 1% steps

- If you're at the 80% percentile on a text, then 80% of students scored below you
- For a sorted dataset $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n$, the p-th percentile corresponds to the value at position

$$k = \frac{p}{100}(n + 1)$$

- For a probability distribution, the percentile can be calculated using the inverse of the CDF:

$$F_X(x) = p$$

$$x = F_X^{-1}(p)$$

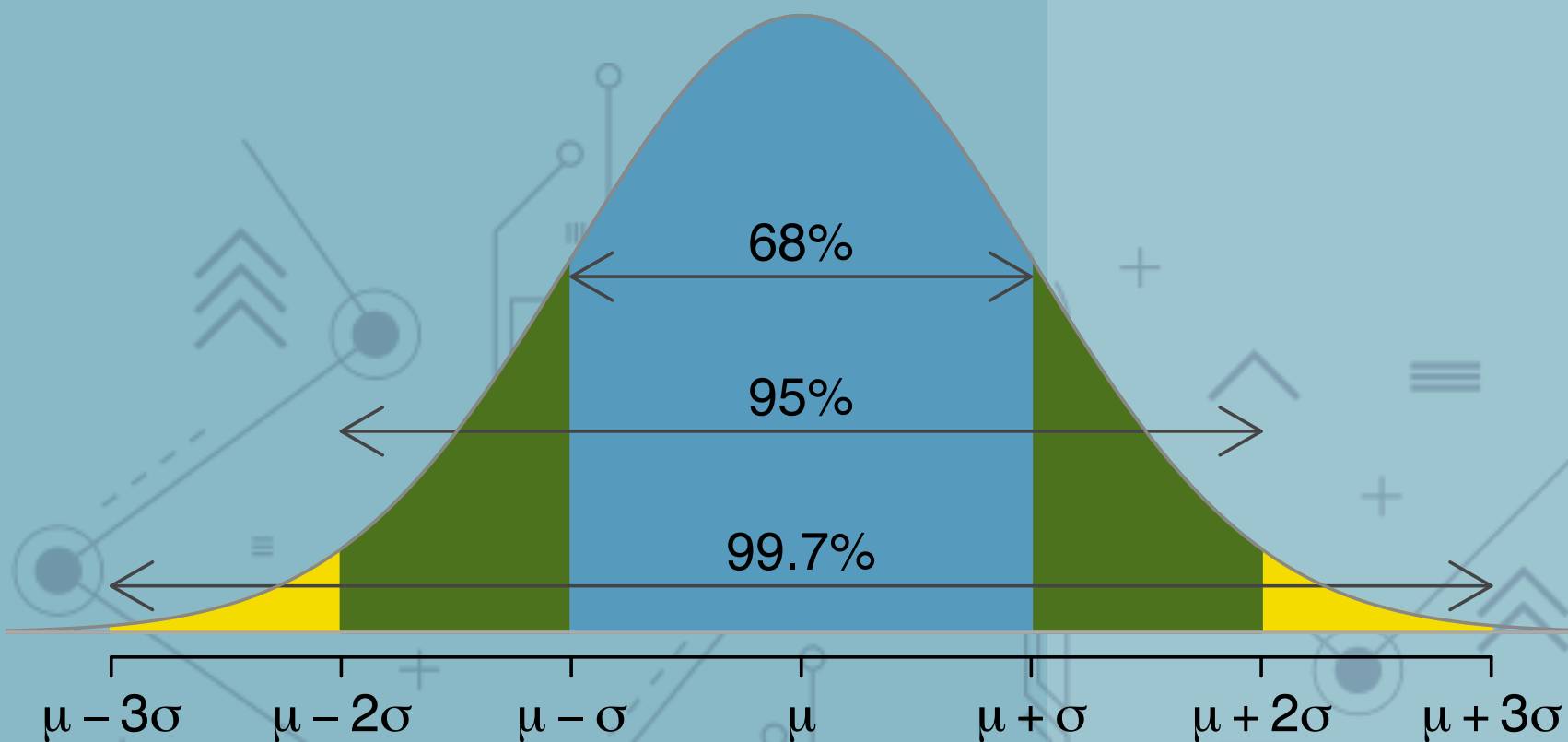
Sigmas

Aka “plot the one sigma errors”

$$N(\mu, \sigma^2)$$

pdf:
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{(x - \mu)^2}{2\sigma^2}\right)$$

cdf:
$$F_X(x) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x - \mu}{\sigma\sqrt{2}}\right) \right]$$



Important: the 68% percentile is only equivalent to 1σ in the case of a Gaussian distribution!

Expectation and Variance

$$N(\mu, \sigma^2)$$

$$\text{E}[X]: \quad f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{(x - \mu)^2}{2\sigma^2}\right)$$

$$\text{cdf:} \quad F_X(x) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x - \mu}{\sigma\sqrt{2}}\right) \right]$$

[TO FINISH]

Bivariate and multi-variate distributions

TO FINISH



Estimators



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- Data is a sample of random variables drawn from some underlying population with a fixed pdf



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- Data is a sample of random variables drawn from some underlying population with a fixed pdf
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Estimator: statistic that can be calculated from a data sample, and approximated the true value of some population pdf parameter. For a population parameter θ , its estimator $\hat{\theta}$ is biased if $E[\hat{\theta}] \neq \theta$

Estimators

A lot of (frequentist) statistics involves calculating estimators, figuring out the statistical distribution they obey, and then comparing the estimators derived from observations with the theoretical distributions they are drawn from.

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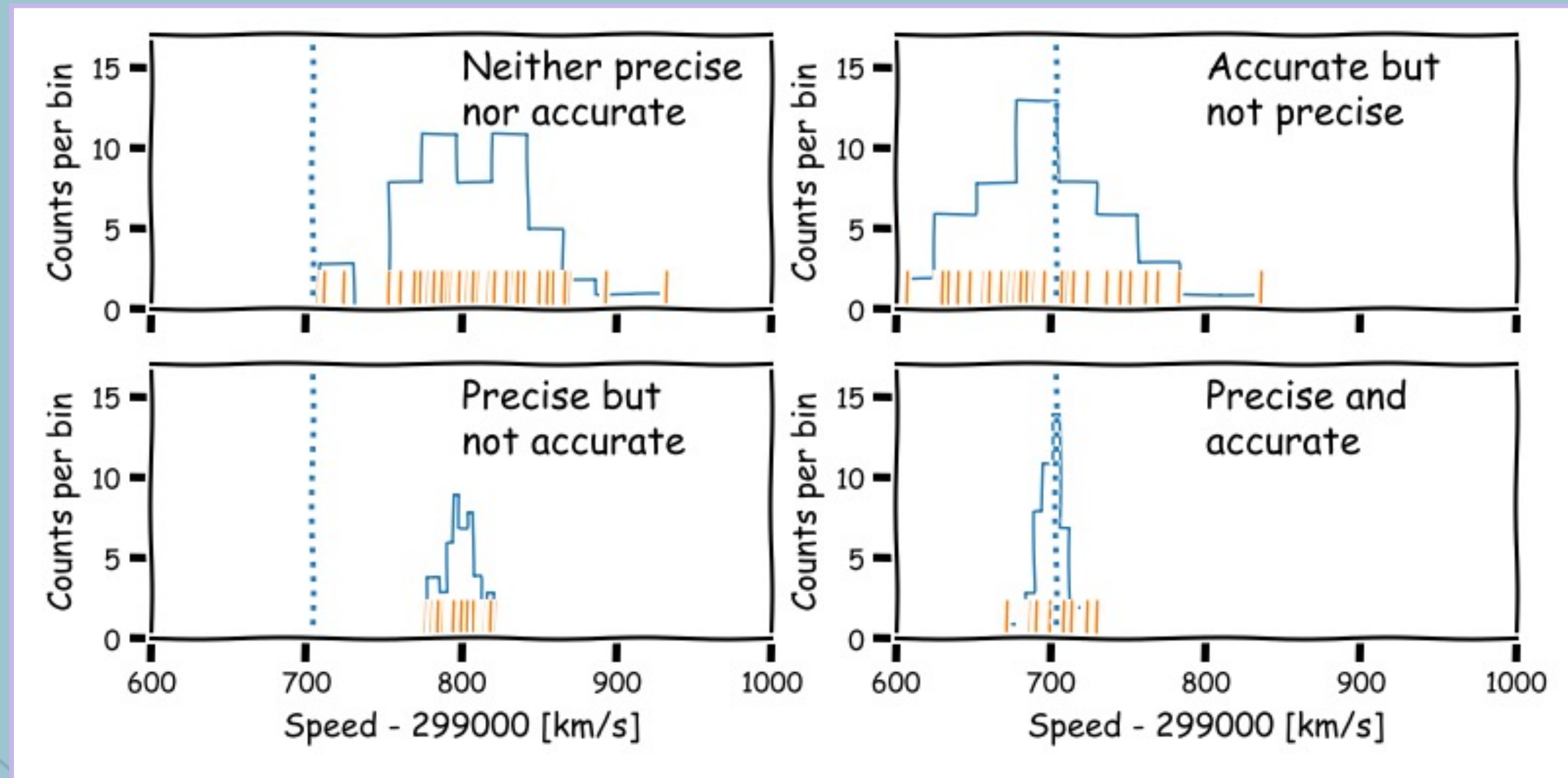
Estimators

Unbiased estimators:

Biased estimators



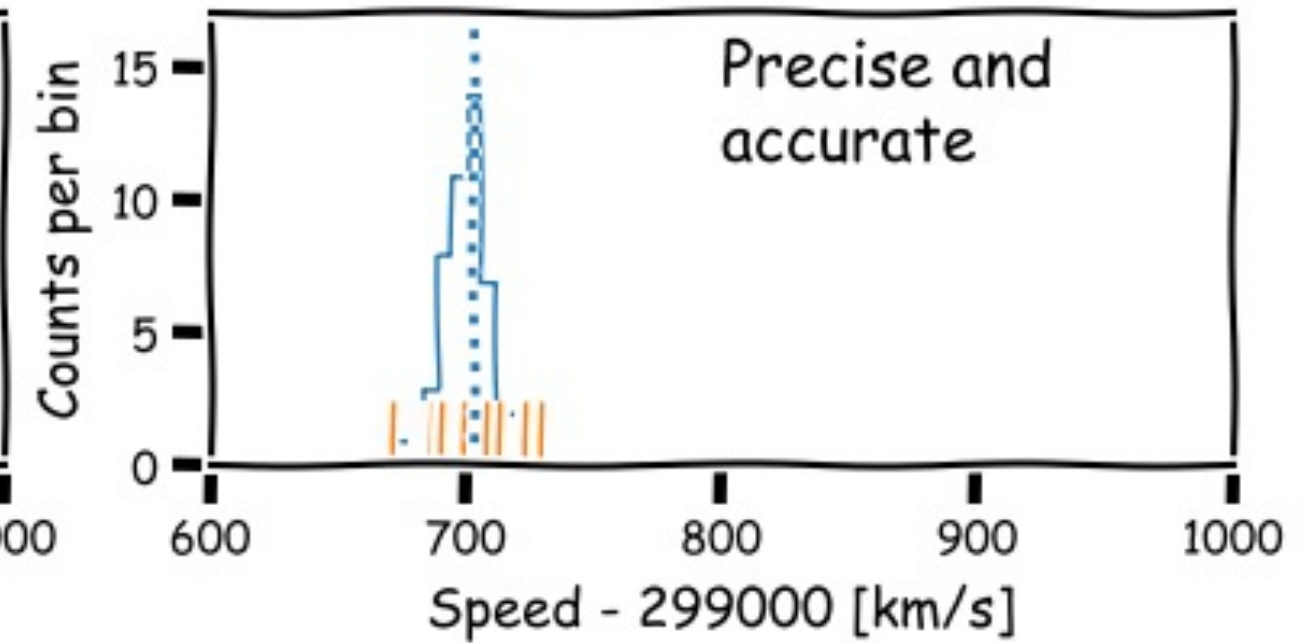
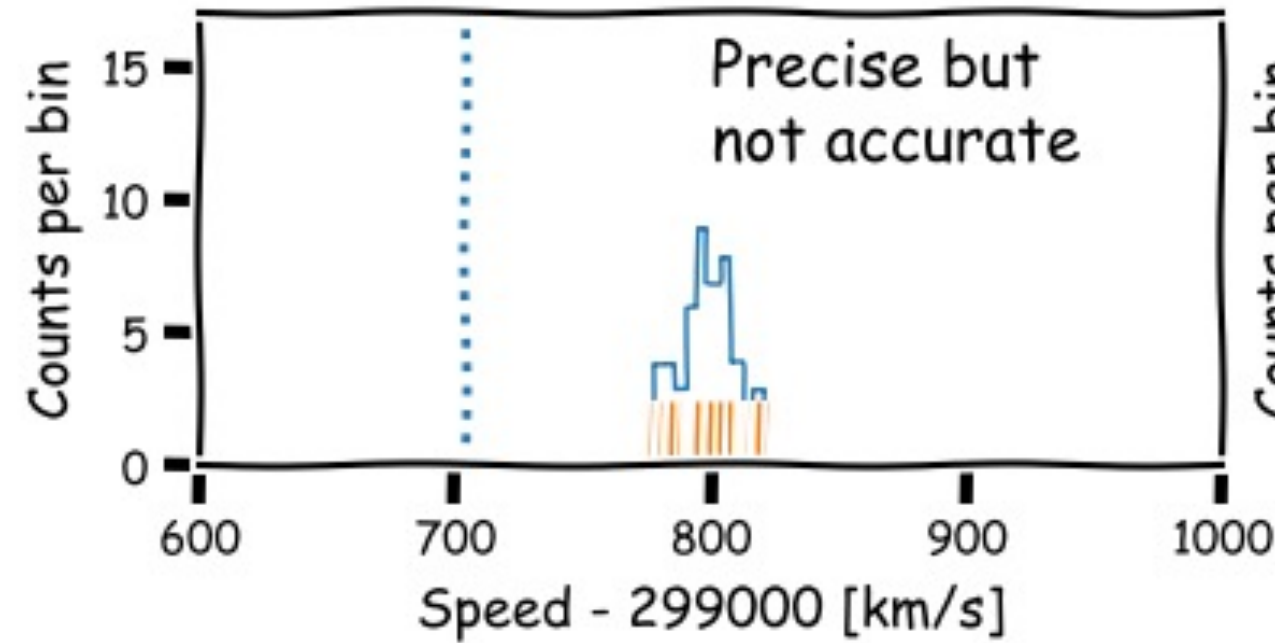
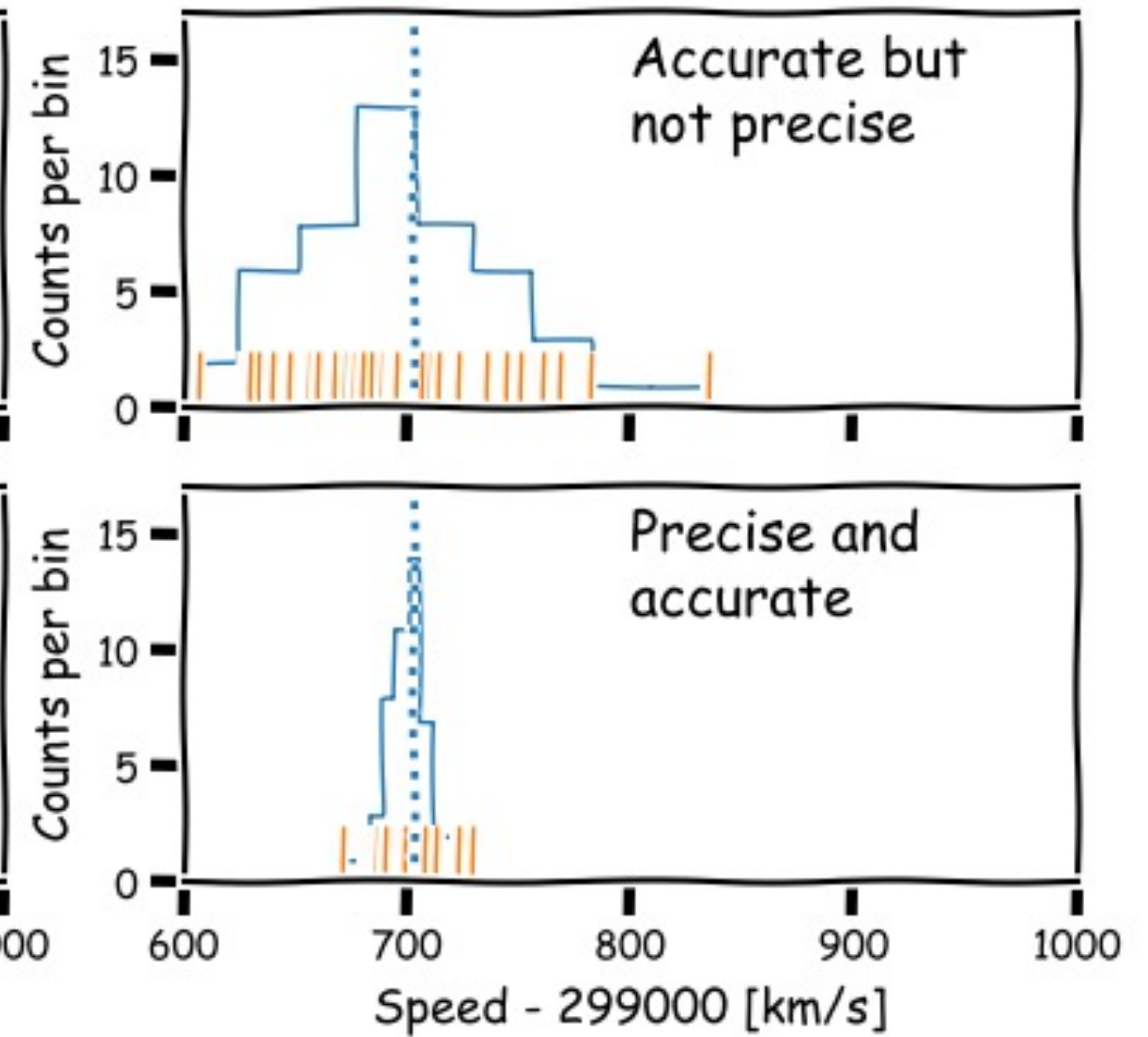
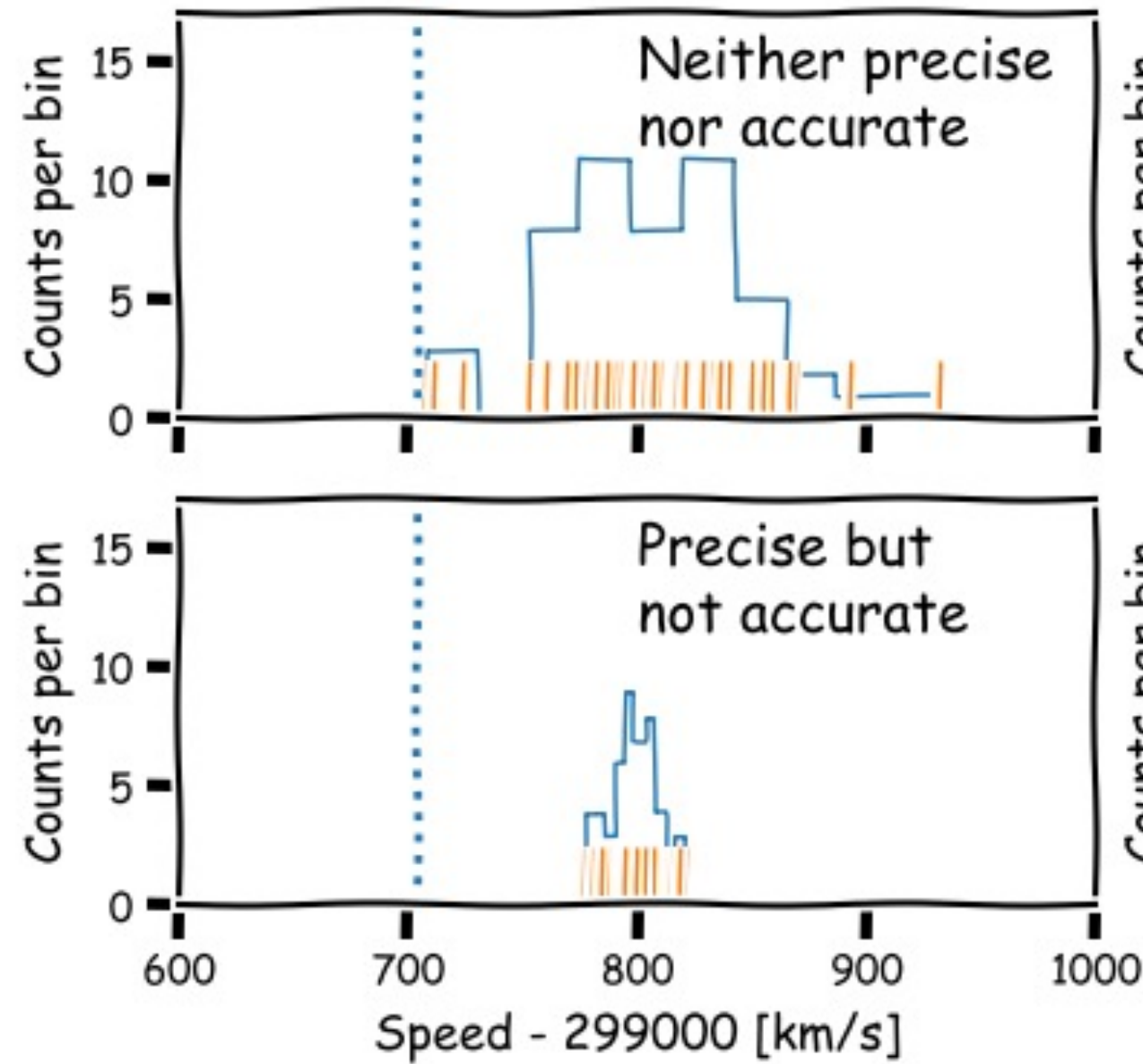
Accuracy versus precision

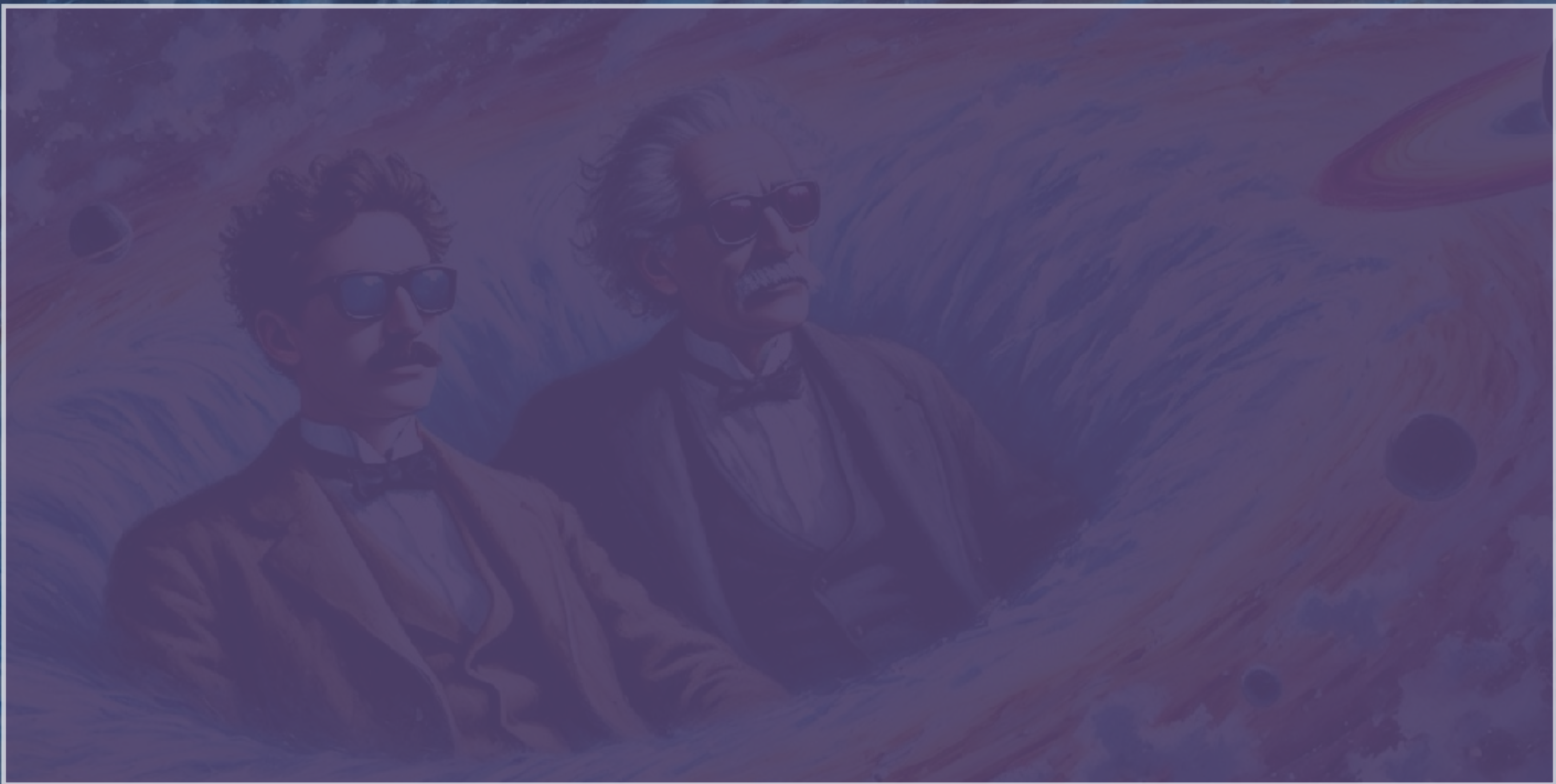


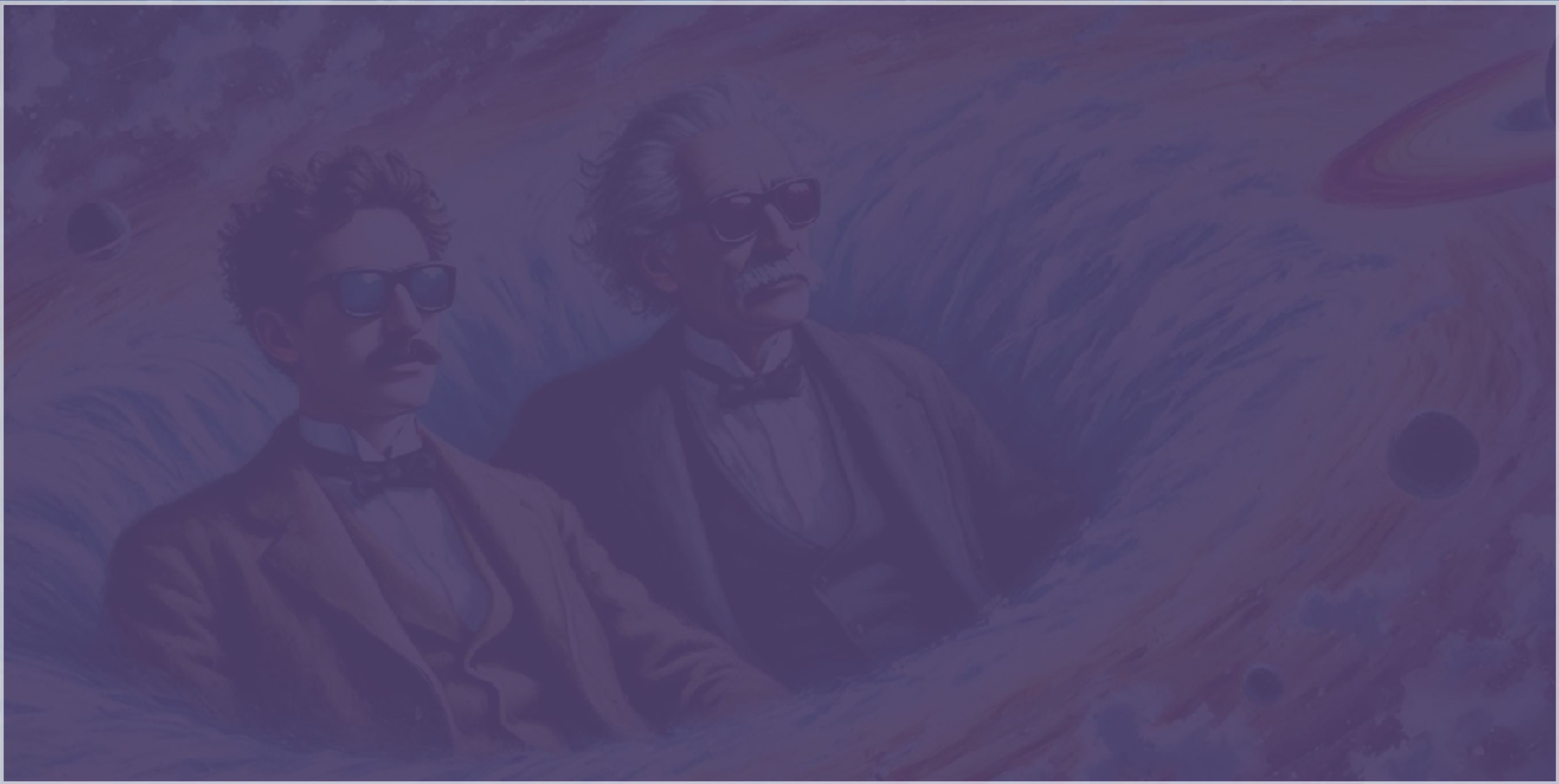
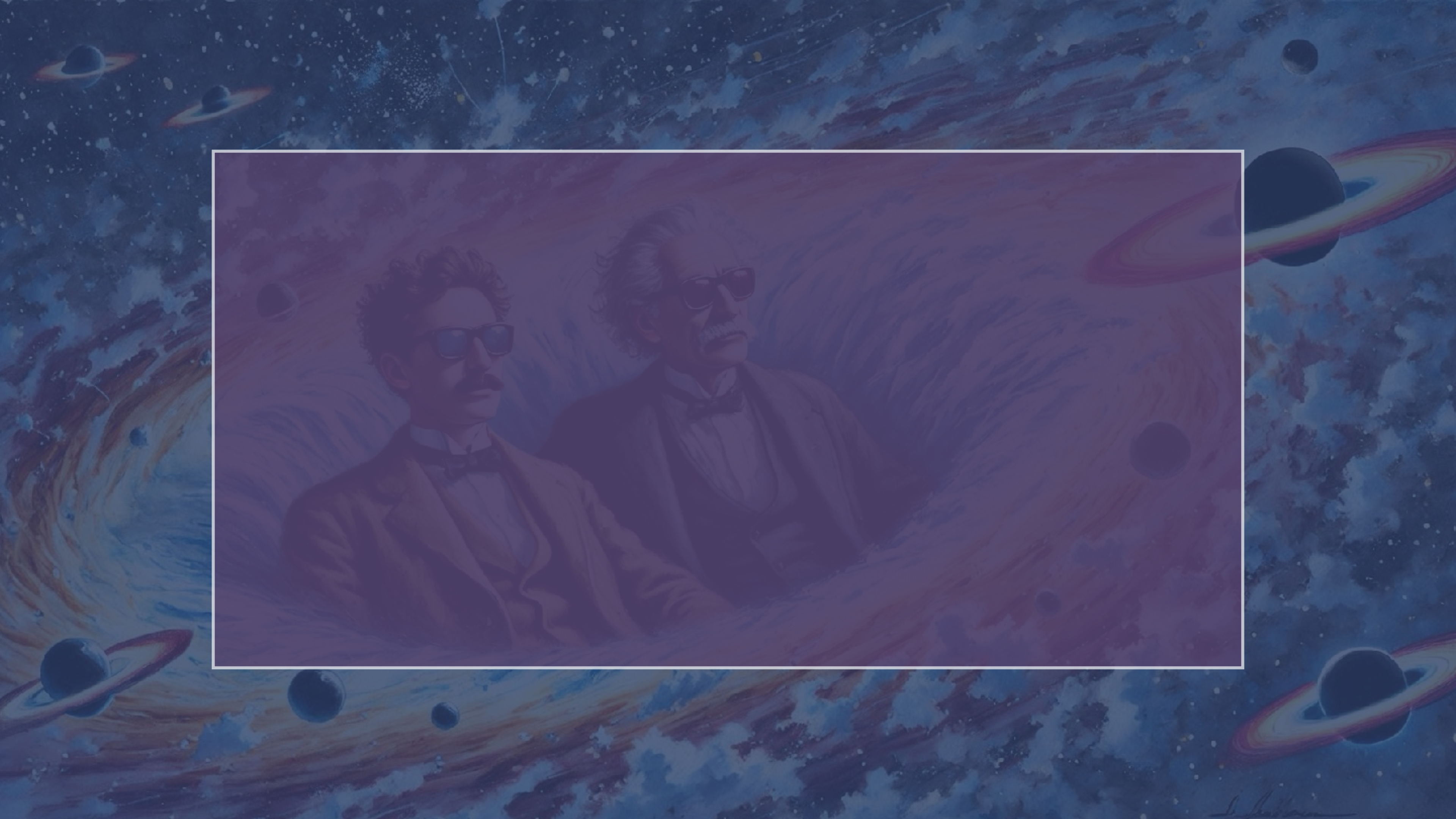
Accuracy versus precision

Accuracy relates to bias or systematic error

Precision of measurements relates to noise or statistics error







Li Huihui